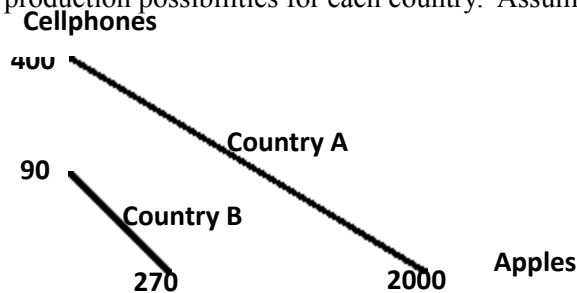


1. The following table shows how many cellphones or apples that a person in Country A or Country B can produce per day. For simplicity assume that all workers within each country have the same skills and that they can move freely between industries within each country, but they cannot move across countries. Furthermore, assume the Country A has 100 workers and Country B has 90 workers.

	Output per person per day	
	Cellphones	Apples
Country A	4	20
Country B	1	3

Use this information to answer the following questions:

- a. Draw the production possibilities frontier (PPF) for each country. Your PPF will show the daily production possibilities for each country. Assume constant opportunity costs (OC).



- b. Which country has the absolute advantage in producing cellphones? Or in producing apples?

Both in Country A; a worker in Country A can make 4 cellphones per day while a worker in Country B can only make 1 Cellphone per day. Similarly, a worker in Country A can make 20 Apples per day while a worker in Country B can make 3 Apples per day.

- c. Calculate the OC of producing a cellphone in each country.

OC of producing a cellphone in each country is 5 Apples for Country A and 3 Apples for Country B.

- d. Calculate the OC of producing an apple in each country.

OC of producing an apple in each country is $\frac{1}{5}$ cellphone for Country A and $\frac{1}{3}$ Apple for Country B.

- e. According to the law of comparative advantage, which product will the Country A export? Which product will the Country B export?

The Country A will export Apples and Country B will export cellphones. The law of comparative advantage says that you should export the good in which you have the lower opportunity cost.

- f. Assume the countries decide to trade with each other. Each specializes in the production of the good in which they have the comparative advantage and exports the good in which they have the comparative advantage. The terms of trade between the two is set at 20 cellphones for 200 apples. After they finish trading, how many cellphones and how many apples will each country be able to consume?

After trading, the Country A will be able to consume 1800 apples ($2000 - 200$) and 20 cellphones. Country B will consume 200 apples and 70 cellphones ($90 - 20$).

- g. Given your answer in part f), will each country's consumption point after trade be inside its PPF? Outside its PPF? Or on its PPF? Provide an explanation.

Country A will consume inside their PPF. We know this because it would be impossible to produce this consumption bundle given their technology and factors of production. For example, the Country A would need 90 workers to produce 1800 apples and another 5 workers to produce 20 cellphones. Yet, they have 100 workers, which is greater than 95. However, Country B would need $200/3$ workers to produce 200 apples and another 70 workers to produce 70 cellphones. But they have 90 workers, smaller than $70 + 200/3$ (approximately 137). Hence, Country B will be at the consumption outside their PPF.

2. Suppose Carpenter A and B both produce Chairs and Desks. The following table shows combinations of the goods that each of them can produce in a day.

Carpenter A		Carpenter B	
Desks	Chairs	Desks	Chairs
0	10	0	20
1	8	1	16
2	6	2	12
3	4	3	8
4	2	4	0
5	0		

- a. Who has an absolute advantage in the production of chairs? What about desks? Explain why.

Carpenter A has an absolute advantage in the production of desks because they can produce more desks than Carpenter B (5 desks vs 4 desks) in a day. Carpenter B has an absolute advantage in the production of chairs because they can produce more chairs (20 chairs vs 10 chairs) in a day.

- b. Calculate the opportunity cost of producing desk in each carpenter. Based on your calculations, which carpenter has a comparative advantage in desk production?

Carpenter A's OC of producing desks is 2 chairs and Carpenter B's OC of producing desks is 4 chairs. Carpenter A has the comparative advantage in desk production.

- c. Calculate the opportunity cost of producing chair in each carpenter. Based on your calculations, which carpenter has a comparative advantage in chair production?

Carpenter A's OC of producing chairs is $\frac{1}{2}$ of desks and Carpenter B's OC of producing chairs is $\frac{1}{4}$ of desks. Carpenter B has the comparative advantage in chair production.

- d. For each of the following terms of trade, determine which carpenter - if any - would agree to trade under these terms.

- 3 desks: 10 chairs (Carpenter A and B would agree to trade)
- 2 desks: 7 chairs (Carpenter A and B would agree to trade)

- 1 desks: 1 chairs (Carpenter B would agree to trade, but Carpenter A would not)
- e. Suppose that Carpenter A is currently producing 2 desks and 6 chairs, and Carpenter B is producing 3 desks and 8 chairs. Explain why carpenters could both be better off (i.e., consume more) if they specialize in producing only one good and then engage in trade. Currently, production is 5 desks and 14 chairs. If Carpenter A specializes in desks and Carpenter B in chairs, total production will be 5 desks and 20 chairs. Those 6 extra chairs can be divided between the two carpenters (e.g., 2 to Carpenter A and 4 to Carpenter B) which would allow both countries to consume outside of their PPF.

** As an extension of those exercise, think about how we can relate above exercises in terms of Marginal Utility, Marginal Rate of Substitution, and Indifference Curve: Marginal Utility vs Marginal Productivity, Marginal rate of Substitution vs Marginal rate of transformation, Indifference Curve vs PPF.